

ELECTRICITY PREDICTION SYSTEM

SUBMITTED BY :

- HARINI SRI S
- JAYASHRE D
- CHARAN ANAND
- VIGNESH K

DECLARATION:

We hereby declare that the Machine Learning Project entitled “**ELECTRICITY PREDICTION SYSTEM**” is our original work which was carried out by four of us in the **FOURTH SEMESTER** during the academic year 2026. Our heartfelt thanks to **DR.MEENALOCHANI** ma’am who helped and guided us throughout the semester and to make this project and to **THE SASTRA UNIVERSITY**, one of the best universities, which gave us the opportunity to create this project.

ABSTRACT:

Electricity usage is an essential function of everyday living, yet accurately predicting the electrical cost of your electricity use is typically not so simple because there are so many factors that influence your electricity use and costs. The intent of this project is to build a machine-learning driven system to determine the electrical cost of an end-user based on certain variables such as the air quality index, utilisation rate and the amount of residents in a given area.

A regression-based approach will be used to analyse the input data to predict electricity costs; however, no interface for actually allowing input will be able to be created until a user-friendly interface is created using Streamlit, which will allow end-users to easily enter their values and receive instant results. The model will be trained on datasets that are pertinent to determining patterns and correlations within the input variable and electrical cost.

The final product will be an efficient and interactive web application that will allow users to obtain accurate and timely predictions with respect to their electricity use and costs..

PROBLEM STATEMENT

Calculating electricity costs manually is difficult because it depends on multiple dynamic factors such as usage patterns, environmental conditions, and population. Existing methods are often static and do not provide real-time predictions.

Therefore, there is a need for an intelligent system that can analyze multiple parameters and provide accurate and quick predictions of electricity costs in an easy-to-use format.

OBJECTIVE :

To develop a user-friendly web-based application for predicting electricity costs using key influencing factors such as utilization rate, air quality index, and resident count. The system aims to apply data-driven techniques to estimate electricity expenses accurately, helping users make informed decisions about energy usage and cost management.

AIM OF THE PROJECT :

The aim of this project is to develop a machine learning-based application that predicts electricity cost using key influencing factors and provides results through a simple and interactive interface.

SYSTEM DESIGN:

The system follows a structured approach:

Step 1: User inputs data (Utilization Rate, AQI, Resident Count)

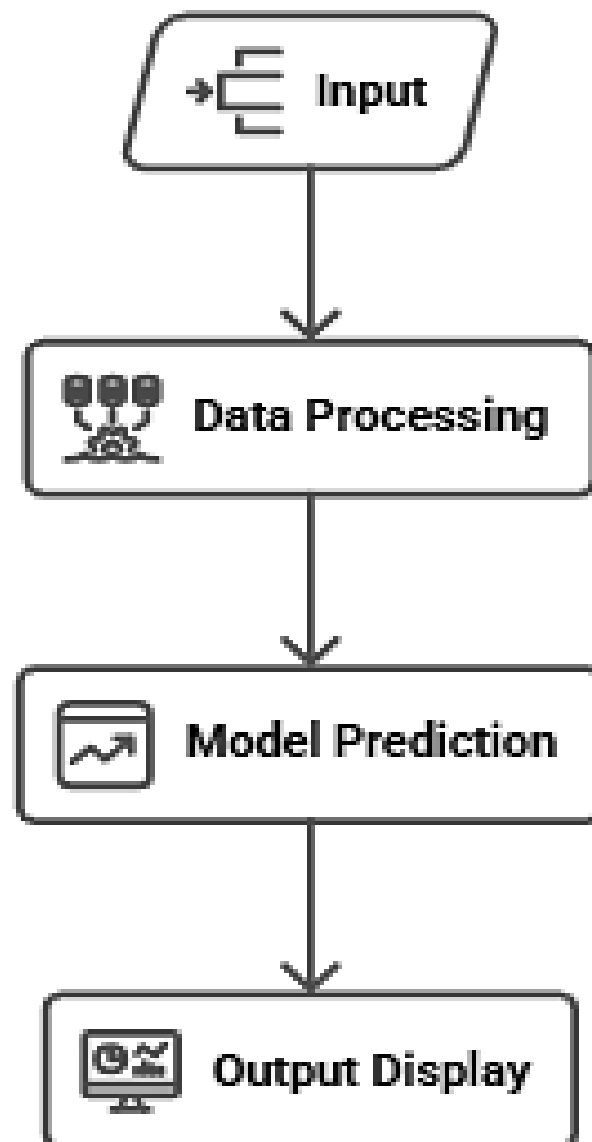
Step 2: Data is processed and passed to the trained model

Step 3: Machine learning model predicts electricity cost

Step 4: Output is displayed on the web interface

Model Used: Linear Regression

FLOW CHART :



TECHNOLOGIES USED :

- **PROGRAMMING LANGUAGE:** PYTHON
- **FRAMEWORK:** STREAMLIT
- **LIBRARIES:** PANDAS, NUMPY, SCIKIT-LEARN

IMPLEMENTATION:

The project is implemented using Python and machine learning libraries. Data is collected and preprocessed to remove inconsistencies. A Linear Regression model is trained using the dataset to learn relationships between input features and electricity cost. The trained model is integrated into a Streamlit application, where users can provide input values using sliders and input fields. The system processes these inputs and displays the predicted electricity cost instantly.

SOURCE CODE :

```
import streamlit as st
import pandas as pd
from sklearn.linear_model import LinearRegression

st.title("☐ Electricity Cost Prediction")

# Load dataset
data = pd.read_csv("electricity_cost_dataset.csv")

# Clean columns
data.columns = data.columns.str.strip().str.lower()
```

```

# Fix names
data = data.rename(columns={
    'air qality index': 'air quality index',
    'issue reolution time': 'issue resolution time'
})

data = data.dropna()

# Features
X = data[['utilisation rate', 'air quality index', 'resident count']]
y = data['electricity cost']

# Train model
model = LinearRegression()
model.fit(X, y)

# Inputs
st.header("Enter Details")

u = st.slider("Utilisation Rate", 0, 100, 50)
a = st.slider("Air Quality Index", 0, 500, 100)
r = st.number_input("Resident Count", min_value=1, value=100)

# Predict
if st.button("Predict ☐"):
    input_data = pd.DataFrame([[u, a, r]],
                              columns=['utilisation rate', 'air quality
index', 'resident count'])

    result = model.predict(input_data)

    st.success(f"Predicted Electricity Cost: ₹ {result[0]:.2f}")

```

DATA SET LINK :

<https://www.kaggle.com/datasets/shalmamuji/electricity-cost-prediction-dataset>

	A	B	C	D	E	F	G	H	I	J
1	site area	structure	water con	recycling	utilisation	air quality	issue reol	resident c	electricity	cost
2	1360	Mixed-use	2519	69	52	188	1	72	1420	
3	4272	Mixed-use	2324	50	76	165	65	261	3298	
4	3592	Mixed-use	2701	20	94	198	39	117	3115	
5	966	Residentia	1000	13	60	74	3	35	1575	
6	4926	Residentia	5990	23	65	32	57	185	4301	
7	3944	Commerci	6960	22	43	194	59	0	3800	
8	3671	Residentia	2803	24	66	149	18	139	3661	
9	3419	Mixed-use	5601	17	61	140	21	38	2538	
10	630	Residentia	1000	20	82	92	56	41	1390	
11	2185	Residentia	2170	32	57	133	52	117	2599	
12	1269	Residentia	1194	16	94	191	8	124	2626	
13	2891	Industrial	5267	78	76	105	25	0	3714	
14	2933	Residentia	4817	41	100	181	32	260	4140	
15	1684	Commerci	2604	53	61	42	70	0	2100	
16	3885	Commerci	2164	12	53	182	40	0	3419	
17	4617	Residentia	8168	53	43	27	29	175	4479	
18	3404	Commerci	2380	78	76	57	69	0	3165	
19	974	Mixed-use	1000	60	30	14	35	30	872	
20	1582	Commerci	2448	70	48	151	33	0	1803	
21	3058	Mixed-use	2994	60	87	106	46	292	2196	
22	2547	Residentia	3903	42	36	113	72	253	3084	
23	3247	Residentia	5123	29	93	89	31	225	3742	
24	1475	Industrial	1000	39	47	189	68	0	2271	
25	2306	Residentia	3978	53	92	192	40	43	2271	

electricity_cost_dataset

Ready

Type here to search

GOOGLE COLLAB LINK :

<https://colab.research.google.com/drive/1tLZdbBUiqX0O3NEJH-AECBJNCmJGHsua?usp=sharing#scrollTo=OiGZnu67ygk4>

OUTPUT :

The system successfully predicts the electricity cost based on user inputs. The interface allows users to adjust parameters and observe how the predicted cost changes.

Example:

Predicted Electricity Cost: ₹3366.72

This demonstrates the effectiveness of the model in providing quick and useful predictions.

Some screenshots of the output are as follows :

Electricity Cost Prediction

Enter Details

Utilisation Rate



Air Quality Index



Resident Count

 - +

Predict

Predicted Electricity Cost: ₹ 39080.339

Electricity Cost Prediction

Enter Details

Utilisation Rate



Air Quality Index



Resident Count

10000

- +

Predict

Predicted Electricity Cost: ₹ 39507.509

FUTURE SCOPE :

- ❖ Add more features like weather and appliance usage
- ❖ Improve accuracy using advanced ML models
- ❖ Integrate real-time data sources
- ❖ Develop a mobile application
- ❖ Add graphical analysis and dashboards

CONCLUSION

The Electricity Cost Prediction system successfully demonstrates the application of machine learning in predicting real-world values. By using key input parameters, the system provides quick and reliable cost estimates through an interactive interface.

This project highlights the importance of data-driven decision-making and serves as a foundation for future improvements in smart energy management systems.